

LAB ASSIGNMENT – 18.1

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BATCH- 02

Task 1 – Weather API Integration

Connect to a Weather API to fetch current weather details for a given city.

Instructions:

- Prompt AI to generate Python code using the requests library.
- Handle errors: invalid city name, API key errors, and network failures.
- Display temperature, humidity, and weather description in a readable format.

Expected Output:

- A working Python script that fetches and displays weather data with proper error handling.

Code :

```
""wethe fetcher use open meteo api and gecoding api ""
```

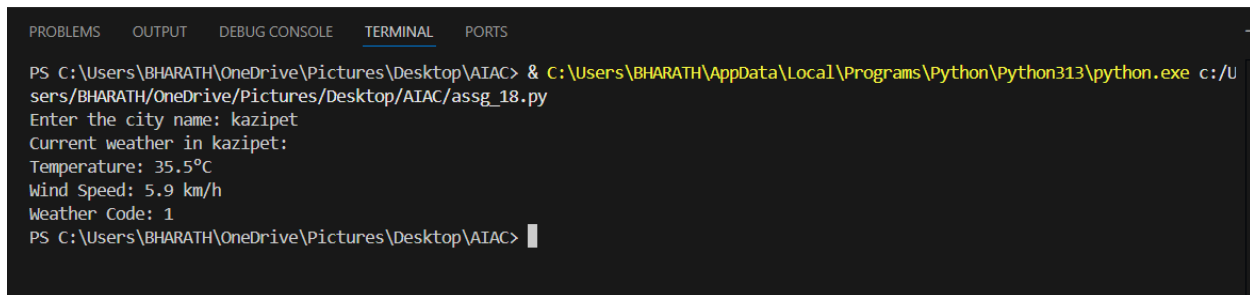
```
import requests
def get_coordinates(city):
    url = f"https://geocoding-api.open-meteo.com/v1/search?name={city}"
    response = requests.get(url)
    data = response.json()
    if "results" in data and len(data["results"]) > 0:
        latitude = data["results"][0]["latitude"]
        longitude = data["results"][0]["longitude"]
        return latitude, longitude
    else:
        raise ValueError("City not found")
def get_weather(latitude, longitude):
```

```

url = f"https://api.open-
meteo.com/v1/forecast?latitude={latitude}&longitude={longitude}&current_weather=true"
response = requests.get(url)
data = response.json()
if "current_weather" in data:
    return data["current_weather"]
else:
    raise ValueError("Weather data not found")
def main():
    city = input("Enter the city name: ")
    try:
        latitude, longitude = get_coordinates(city)
        weather = get_weather(latitude, longitude)
        print(f"Current weather in {city}:")
        print(f"Temperature: {weather['temperature']}°C")
        print(f"Wind Speed: {weather['windspeed']} km/h")
        print(f"Weather Code: {weather['weathercode']}")
    except ValueError as e:
        print(e)
if __name__ == "__main__": main()

```

Output :



```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\BHARATH\OneDrive\Pictures\Desktop\AIAC> & C:\Users\BHARATH\AppData\Local\Programs\Python\Python313\python.exe c:/U
sers/BHARATH/OneDrive/Pictures/Desktop/AIAC/assg_18.py
Enter the city name: kazipet
Current weather in kazipet:
Temperature: 35.5°C
Wind Speed: 5.9 km/h
Weather Code: 1
PS C:\Users\BHARATH\OneDrive\Pictures\Desktop\AIAC>

```

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API downtime.
- Show conversion results clearly in tabular format.

Expected Output:

- A script that converts INR to multiple currencies with graceful

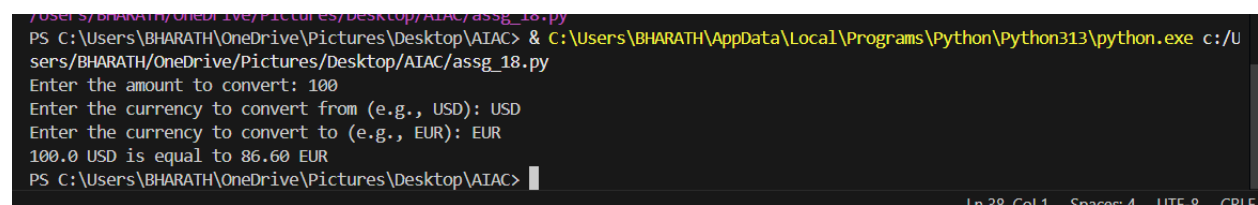
error handling.

Code:

"""desing a currency exchanger that can concert into different currency and vice versa use
currency exchange api"""

```
import requests
def get_exchange_rate(from_currency, to_currency):
    url = f"https://api.exchangerate-api.com/v4/latest/{from_currency}"
    response = requests.get(url)
    data = response.json()
    if "rates" in data and to_currency in data["rates"]:
        return data["rates"][to_currency]
    else:
        raise ValueError("Currency not found")
def convert_currency(amount, from_currency, to_currency):
    exchange_rate = get_exchange_rate(from_currency, to_currency)
    return amount * exchange_rate
def main():
    amount = float(input("Enter the amount to convert: "))
    from_currency = input("Enter the currency to convert from (e.g., USD): ")
    to_currency = input("Enter the currency to convert to (e.g., EUR): ")
    try:
        converted_amount = convert_currency(amount, from_currency, to_currency)
        print(f"{amount} {from_currency} is equal to {converted_amount:.2f} {to_currency}")
    except ValueError as e:
        print(e)
if __name__ == "__main__": main()
```

Output:



```
PS C:\Users\BHARATH\OneDrive\Pictures\Desktop\AIAC> & C:\Users\BHARATH\AppData\Local\Programs\Python\Python313\python.exe c:/Users/BHARATH/OneDrive/Pictures/Desktop/AIAC/assg_18.py
Enter the amount to convert: 100
Enter the currency to convert from (e.g., USD): USD
Enter the currency to convert to (e.g., EUR): EUR
100.0 USD is equal to 86.60 EUR
PS C:\Users\BHARATH\OneDrive\Pictures\Desktop\AIAC>
```

Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

Instructions:

- Prompt AI to write a Python function to send GET requests to GitHub API.
- Handle rate limit errors, 404 (repo not found), and invalid input.
- Display repository name, description, stars, forks, and open issues.

Expected Output:

- A Python program that retrieves and displays GitHub repository information with robust error handling.

Code:

```
""write a python function to send GET request to github api ""
```

```
import requests

def get_github_user(username):
    url = f"https://api.github.com/users/{username}"
    response = requests.get(url)
    if response.status_code == 200:
        data = response.json()
        return {
            "login": data["login"],
            "name": data["name"],
            "public_repos": data["public_repos"],
            "followers": data["followers"],
            "following": data["following"]
        }
    else:
        raise ValueError("User not found")

def main():
    username = input("Enter the GitHub username: ")
    try:
        user_info = get_github_user(username)
        print(f"Login: {user_info['login']}")
        print(f"Name: {user_info['name']}")
        print(f"Public Repositories: {user_info['public_repos']}")
        print(f"Followers: {user_info['followers']}")
        print(f"Following: {user_info['following']}")
    except ValueError as e:
        print(e)
```

```
if __name__ == "__main__": main()
```

Output:

```
PS C:\Users\BHARATH\OneDrive\Pictures\Desktop\AIAC> & C:\Users\BHARATH\AppData\Local\Programs\Python\Python313\python.exe c:/Users/BHARATH/OneDrive/Pictures/Desktop/AIAC/assg_18.py
Enter the GitHub username: sadff
Login: sadff
Name: None
Public Repositories: 1
Followers: 2
Following: 0
PS C:\Users\BHARATH\OneDrive\Pictures\Desktop\AIAC> |
```

Task 4 – Real-Time Application: News Headlines Aggregator

Scenario: Build a News Aggregator using a free News API.

Requirements:

- Fetch top 5 headlines for a given category (sports, technology, health).
- Handle errors like invalid category, missing API key, and request failures.
- Display headlines in a user-friendly numbered list format.
- Include a retry mechanism if the first request fails.

Expected Output:

- A working Python script that retrieves and displays news headlines with proper error handling

Code:

```
"""
```

News Headlines Fetcher

use newsapi.org

```
"""
```

```
import requests
```

```
def get_news_headlines(category, api_key):
```

```
    """Fetch top 5 news headlines for a given category using the NewsAPI"""
```

```
    url = f"https://newsapi.org/v2/top-headlines?category={category}&apiKey={api_key}"
```

```
    response = requests.get(url)
```

```
    if response.status_code == 200:
```

```
        data = response.json()
```

```
        if "articles" in data:
```

```

        return [article["title"] for article in data["articles"][:5]]
    else:
        raise ValueError("No articles found")

elif response.status_code == 401:
    raise ValueError("Invalid API key")

elif response.status_code == 400:
    raise ValueError("Invalid category")

else:
    raise ValueError(f"Error: {response.status_code}")

def main():
    api_key = input("Enter your News API key: ").strip()
    category = input("Enter category (business, entertainment, health, science, sports,
technology): ").strip().lower()

    try:
        headlines = get_news_headlines(category, api_key)
        print("\nTop 5 Headlines:\n")
        for i, title in enumerate(headlines, 1):
            print(f"{i}. {title}")

    except ValueError as e:
        print("Error:", e)

if __name__ == "__main__":
    main()

```

Output:

```

sers/BHARATH/OneDrive/Pictures/Desktop/AIAC/assg_18.py
Enter your News API key: 71b8ebd4f2f2456191db37f7faf6413a
Enter category (business, entertainment, health, science, sports, technology): sports

Top 5 Headlines:

1. Fantasy Baseball SP Roundup 3/25: Dud Webb - Pitcher List
2. Joel Berry reacts to Hubert Davis firing, reveals key factor in decision - On3
3. NC State Basketball Hoop Scoop: Intel on Will Wade, Boo Corrigan Meeting - 247Sports
4. Player Grades: Lakers vs. Pacers - Silver Screen and Roll
5. First SF Giants at-bats ruined by Netflix's aggressive ad insertion - SFGATE
PS C:\Users\BHARATH\OneDrive\Pictures\Desktop\AIAC> █

```